# ESKOOLIA SCHOOL ERP

## Student Info Module

User Acceptance Testing (UAT) Report

Comprehensive End-to-End Testing
with Improvement Recommendations & AI Integration Strategy

Date: March 28, 2026
Application URL: http://187.127.137.154/
Tester: Claude (AI-Assisted UAT)
**Total: 71 Test Cases | 42 Pass | 16 Fail | 10 Critical | 3 Warnings**

# 1. Executive Summary

This report presents findings from a comprehensive UAT exercise on the Student Info module of Eskoolia School ERP. The module was tested across 8 sub-modules covering students, categories, groups, guardians, promotions, documents, transfers, multi-class records, and attendance. A total of 71 test cases were executed spanning CRUD operations, input validation, security/permissions, and UI/UX evaluation across 15 UI pages.

## 1.1 Test Results Summary

| Category | PASS | FAIL | CRITICAL | WARN |
|---|---|---|---|---|
| **Student CRUD** | 10 | 2 | 0 | 0 |
| **Category CRUD** | 4 | 1 | 0 | 0 |
| **Group CRUD** | 3 | 0 | 0 | 0 |
| **Sub-modules (Guardian/Promo/Doc/Transfer/Multi)** | 6 | 2 | 0 | 0 |
| **Attendance Operations** | 3 | 2 | 0 | 0 |
| **Input Validation** | 4 | 3 | 0 | 1 |
| **Security & Permissions** | 0 | 0 | 10 | 0 |
| **UI/UX Testing** | 12 | 6 | 0 | 2 |
| **TOTAL (71 tests)** | **42** | **16** | **10** | **3** |

## 1.2 Critical Findings

1. **ZERO Permission Enforcement (P0):** All 10 security tests returned CRITICAL. A teacher with 0 permissions can READ all 797 student records, CREATE new students, DELETE student records, CREATE categories, and MARK attendance for any student. This is the same systemic vulnerability found in Role Permission and Admin Section modules.
2. **Duplicate Key Returns 500 (P1):** Three separate duplicate scenarios return 500: duplicate admission_no on students, duplicate category name, and duplicate attendance (same student+date). All should return 400 with validation messages. Pattern is systemic across the entire ERP.
3. **XSS Vulnerability - HTML Stored in Names (P1):** HTML tags like Bold and <script> are accepted and stored in student first_name fields. The Student List UI renders these without sanitization. If the frontend does not escape HTML, this enables stored XSS attacks where malicious scripts execute for every user viewing the student list.
4. **Student Export Shows Only 25 of 797 (P1):** The Student Export page shows 'Total students ready for export: 25' when there are 797 students in the system. The export function only processes the first page of paginated results, silently dropping 97% of student data.
5. **Section Dropdown Not Filtered by Class (P2):** On Multi Class, Delete Record, Disabled Students, and Promote pages, the Section dropdown shows ALL 25 sections from ALL 12

classes instead of filtering by the selected class. Users see 12 'A' options and 12 'B' options with no way to distinguish which belongs to which class.

# 2. Module Architecture

## 2.1 API Endpoints Tested

Base URL: http://187.127.137.154/api/v1/students/

| Endpoint | Methods Tested | Status |
|---|---|---|
| students/students/ | GET, POST, PATCH, PUT, DELETE, OPTIONS | Functional (validation gaps) |
| students/categories/ | GET, POST, PATCH, DELETE | Functional (duplicate 500) |
| students/groups/ | GET, POST, DELETE | Fully Functional |
| students/guardians/ | GET | Functional (0 records) |
| students/promotions/ | GET | Functional (2 records) |
| students/documents/ | GET | Functional (0 records) |
| students/transfers/ | GET | Functional (0 records) |
| students/multi-class-records/ | GET | Functional (2 records) |
| attendance/student-attendance/ | GET, POST | Functional (duplicate 500) |
| attendance/subject-attendance/ | GET | 404 - Not Implemented |
| attendance/student/ | GET | 404 - Not in REST router |

## 2.2 UI Pages Audited (15 pages)

| Page URL | Finding |
|---|---|
| /students/category | Simple form + list, 11 categories, functional |
| /students/add | Comprehensive form with 12+ fields, guardian section |
| /students/list | Search, Active toggle, table - XSS data rendered unsafe |
| /students/multi-class | Section dropdown not filtered by class |
| /students/delete-record | Search + class/section filter, same dropdown bug |
| /students/unassigned | Search field only, minimal UI |
| /attendance/student | Year/class/section/date filter, monthly report gen |
| /attendance/student/import | Download sample, file upload, class/section filter |
| /students/group | Name + description form, 2 groups |
| /students/promote | Target year dropdown shows only past years |
| /students/disabled | Table with correct columns, section dropdown bug |
| /attendance/subject | Class/section/subject selector |
| /attendance/subject-report | Class + month selector (01-12) |
| /students/export | Shows only 25 of 797 students for export |

| | |
|---|---|
| /students/sms-sending-time | Placeholder - not implemented |

# 3. Detailed Test Results

## 3.1 Student CRUD Operations

| ID | Test Case | Result / Notes | Status | Severity |
|---|---|---|---|---|
| **TC-S01** | Create student (POST /students/students/) | 201 Created with valid class/section IDs | **PASS** | P3 |
| **TC-S02** | Read student list (GET paginated) | 200 OK, count=797, paginated results | **PASS** | P3 |
| **TC-S03** | Read single student (GET /:id) | 200 OK, full student object with 24 fields | **PASS** | P3 |
| **TC-S04** | Update student (PATCH) | 200 OK, first_name updated correctly | **PASS** | P3 |
| **TC-S05** | Update student (PUT) | 200 OK, full replace successful | **PASS** | P3 |
| **TC-S06** | Delete student (DELETE) | 204 No Content | **PASS** | P3 |
| **TC-S07** | Delete student with FK constraints | 500 - students with attendance/marks/fees cannot be deleted | **FAIL** | P1 |
| **TC-S08** | Create with empty body | 400 validation error returned | **PASS** | P3 |
| **TC-S09** | Create with missing admission_no | 400 required field validation | **PASS** | P3 |
| **TC-S10** | Create with invalid class ID | 400 'object does not exist' error | **PASS** | P3 |
| **TC-S11** | Duplicate admission_no | 500 instead of 400 validation error | **FAIL** | P1 |
| **TC-S12** | OPTIONS on students endpoint | 200 with full field schema (POST actions) | **PASS** | P3 |

## 3.2 Category CRUD Operations

| ID | Test Case | Result / Notes | Status | Severity |
|---|---|---|---|---|
| **TC-C01** | Read categories list | 200 OK, 11 categories returned | **PASS** | P3 |
| **TC-C02** | Create category | 201 Created with ID | **PASS** | P3 |
| **TC-C03** | Update category (PATCH) | 200 OK, name updated | **PASS** | P3 |
| **TC-C04** | Delete category | 204 No Content | **PASS** | P3 |
| **TC-C05** | Duplicate category name | 500 instead of 400 - no unique constraint handling | **FAIL** | P1 |

## 3.3 Group CRUD Operations

| ID | Test Case | Result / Notes | Status | Severity |
|---|---|---|---|---|
| **TC-G01** | Read groups list | 200 OK, 2 groups with students_count | **PASS** | P3 |
| **TC-G02** | Create group | 201 Created | **PASS** | P3 |
| **TC-G03** | Delete group | 204 No Content | **PASS** | P3 |

## 3.4 Sub-module Endpoints

| ID | Test Case | Result / Notes | Status | Severity |
|---|---|---|---|---|
| **TC-M01** | Read guardians list | 200 OK, 0 records (no guardians linked) | **PASS** | P3 |
| **TC-M02** | Read promotions list | 200 OK, 2 promotion records | **PASS** | P3 |
| **TC-M03** | Read documents list | 200 OK, 0 records | **PASS** | P3 |
| **TC-M04** | Read transfers list | 200 OK, 0 records | **PASS** | P3 |
| **TC-M05** | Read multi-class records | 200 OK, 2 records | **PASS** | P3 |
| **TC-M06** | Read student attendance (student-attendance/) | 200 OK, 15862 records, paginated | **PASS** | P3 |
| **TC-M07** | Subject attendance API | 404 Not Found - endpoint not registered | **FAIL** | P2 |
| **TC-M08** | Attendance import API | 404 Not Found - not in REST router | **FAIL** | P2 |

## 3.5 Attendance Operations

| ID | Test Case | Result / Notes | Status | Severity |
|---|---|---|---|---|
| **TC-AT01** | Create attendance record | 201 Created with student/class/section/date/type | **PASS** | P3 |
| **TC-AT02** | Invalid attendance_type ('X') | 400 rejected correctly | **PASS** | P3 |
| **TC-AT03** | Invalid student ID (99999) | 400 rejected correctly | **PASS** | P3 |
| **TC-AT04** | Duplicate attendance (same student+date) | 500 instead of 400 - missing unique_together handling | **FAIL** | P1 |
| **TC-AT05** | Future attendance date (2027-01-01) | 201 accepted - no date boundary validation | **FAIL** | P2 |

## 3.6 Input Validation

| ID | Test Case | Result / Notes | Status | Severity |
|---|---|---|---|---|
| **TC-V01** | Invalid gender value | 400 rejected correctly | **PASS** | P3 |
| **TC-V02** | Invalid date format | 400 rejected correctly | **PASS** | P3 |
| **TC-V03** | admission_no > 40 chars (max_length) | 400 rejected correctly | **PASS** | P3 |
| **TC-V04** | Non-existent class ID | 400 'does not exist' error | **PASS** | P3 |
| **TC-V05** | Future date of birth (2030-01-01) | 201 accepted - no DOB range validation | **FAIL** | P2 |
| **TC-V06** | HTML tags in first_name | 201 accepted, stored as-is - XSS risk | **FAIL** | P1 |

| | | | | |
|---|---|---|---|---|
| **TC-V07** | SQL injection in first_name | 201 accepted (ORM protects execution, but stores raw payload) | **WARNING** | P2 |
| **TC-V08** | Spelling error in category 'Scholership' | Data integrity - should be 'Scholarship' | **FAIL** | P3 |

## 3.7 Security & Permission Tests

All security tests used a Teacher account (username: demo, password: 123456, user_id=3) with ZERO permissions assigned. Every single test returned CRITICAL.

| ID | Test Case | Result / Notes | Status | Severity |
|---|---|---|---|---|
| **TC-SEC01** | Teacher reads all students (0 perms) | 200 OK - full read access to 799 student records | **CRITICAL** | P0 |
| **TC-SEC02** | Teacher reads categories (0 perms) | 200 OK - full access | **CRITICAL** | P0 |
| **TC-SEC03** | Teacher reads groups (0 perms) | 200 OK - full access | **CRITICAL** | P0 |
| **TC-SEC04** | Teacher reads all attendance (0 perms) | 200 OK - full access to 15862+ records | **CRITICAL** | P0 |
| **TC-SEC05** | Teacher creates student (0 perms) | 201 Created - can add students without permission | **CRITICAL** | P0 |
| **TC-SEC06** | Teacher deletes student (0 perms) | 204 - can delete student records without permission | **CRITICAL** | P0 |
| **TC-SEC07** | Teacher creates category (0 perms) | 201 - can create categories without permission | **CRITICAL** | P0 |
| **TC-SEC08** | Teacher creates attendance (0 perms) | 201 - can mark attendance without permission | **CRITICAL** | P0 |
| **TC-SEC09** | Teacher reads promotions (0 perms) | 200 OK - full access | **CRITICAL** | P0 |
| **TC-SEC10** | Teacher reads multi-class (0 perms) | 200 OK - full access | **CRITICAL** | P0 |

## 3.8 UI/UX Testing

| ID | Test Case | Result / Notes | Status | Severity |
|---|---|---|---|---|
| **TC-U01** | Student Category page | Simple form + list, 11 categories, Edit/Delete buttons | **PASS** | P3 |
| **TC-U02** | Add Student page | Comprehensive form: admission, name, DOB, gender, blood group, religion, category, class, section, guardian | **PASS** | P3 |
| **TC-U03** | Student List page | Search by name/admission/roll, Active Only toggle, Refresh button, table with status column | **PASS** | P3 |
| **TC-U04** | Student List - XSS stored data visible | HTML/script tags from test records rendered in table - potential XSS if not escaped | **FAIL** | P1 |
| **TC-U05** | Multi Class Student page | Class/section dropdowns, Add/Load/Update per student | **PASS** | P3 |
| **TC-U06** | Section dropdown not filtered by class | All sections (12xA, 12xB, 1xC) shown regardless of selected class | **FAIL** | P2 |

| | | | | |
|---|---|---|---|---|
| **TC-U07** | Delete Student Record page | Search + class/section filter + checkbox select | **PASS** | P3 |
| **TC-U08** | Unassigned Student page | Search field only, shows students without class assignment | **PASS** | P3 |
| **TC-U09** | Student Attendance page | Academic year, class/section/date filter, Import button, monthly report generator | **PASS** | P3 |
| **TC-U10** | Attendance Import page | Download Sample File link, class/section/date, file upload | **PASS** | P3 |
| **TC-U11** | Student Group page | Name + description form, 2 groups listed | **PASS** | P3 |
| **TC-U12** | Student Promote page | Current vs target year/class/section, but target year shows only past years | **FAIL** | P2 |
| **TC-U13** | Disabled Students page | Class/section filter, table columns correct, section dropdown bug present | **WARNING** | P2 |
| **TC-U14** | Subject Wise Attendance page | Class/section/subject selector, Search button | **PASS** | P3 |
| **TC-U15** | Subject Attendance Report page | Class selector + month (01-12), Search button | **PASS** | P3 |
| **TC-U16** | Student Export page | Export to CSV/PDF buttons, shows only 25 of 797 students | **FAIL** | P1 |
| **TC-U17** | SMS Sending Time page | Placeholder page - not implemented, shows legacy parity message | **FAIL** | P2 |
| **TC-U18** | Class 2 missing value attribute in dropdowns | Option for Class 2 has no value attribute in multiple pages | **FAIL** | P2 |
| **TC-U19** | Breadcrumb navigation | Present on all pages: Dashboard > Student Information > [Page] | **PASS** | P3 |
| **TC-U20** | No pagination on Student List UI | All records load at once (client-side), no page controls | **WARNING** | P2 |

# 4. Bug Summary & Priority Matrix

## 4.1 P0 - Critical (Blocker)

1. **BUG-S001: Zero Permission Enforcement on Student APIs**

Any authenticated user (teacher, driver, student) can READ all student records, CREATE new students, DELETE students, CREATE categories/groups, and MARK attendance. The custom permission model from access-control is completely unenforced on student and attendance APIs.

Impact: Complete PII breach. Student personal data (name, DOB, gender, blood group, guardian info, address) accessible to all users. A disgruntled teacher could delete all student records or mark false attendance.

Fix: Same as Admin Section - implement custom DRF permission class checking role permissions from access_control tables. Apply globally.

## 4.2 P1 - High

1. **BUG-S002: Duplicate admission_no Returns 500**

Creating a student with an existing admission_no returns HTTP 500. The IntegrityError from the database unique constraint bubbles up unhandled. Should return 400 with 'admission_no already exists' message.

2. **BUG-S003: Duplicate Category Name Returns 500**

Same pattern as BUG-S002. Adding unique_together or serializer-level validation would fix both.

3. **BUG-S004: Duplicate Attendance Returns 500**

Creating attendance for the same student+date combination returns 500 instead of 400. Should enforce unique_together on (student, attendance_date) and return a clear validation message.

4. **BUG-S005: XSS - HTML Tags Stored and Rendered in Student Names**

The API accepts HTML/script tags in first_name and last_name fields. The Student List page renders these without HTML escaping. This enables stored XSS attacks where any user creating a student with malicious name content can execute scripts on every user viewing the student list.

Fix: Server-side input sanitization (strip HTML tags from name fields) AND frontend output escaping (React auto-escapes by default unless using dangerouslySetInnerHTML).

5. **BUG-S006: Student Export Only Exports 25 of 797 Records**

The export page only processes the first page of paginated API results. It shows 'Total students ready for export: 25' when there are 797 students. Schools relying on this for data backup or reporting would lose 97% of their data.

Fix: The export function must fetch ALL pages or use a dedicated export API endpoint that returns unpaginated results.

6. **BUG-S007: FK Constraint Error on Student Delete (500)**

Deleting a student who has attendance, marks, or fee records returns 500 Internal Server Error. Should return 400/409 with message: 'Cannot delete student with existing attendance/marks/fee records. Archive the student instead.'

## 4.3 P2 - Medium

1. **BUG-S008: Section Dropdown Shows All Sections (Not Filtered by Class)**

On Multi Class, Delete Record, Disabled Students, and Promote pages, the section dropdown loads ALL 25 sections from every class. Users see 12 identical 'A' options and 12 identical 'B' options with no way to distinguish them. Fix: Filter sections by selected class using dependent dropdown.

2. **BUG-S009: Class 2 Missing value Attribute in Dropdowns**

The option element for Class 2 in all class dropdowns has no value attribute, while all other classes (1, 3-12) do. This could cause Class 2 selections to fail silently.

3. **BUG-S010: Future DOB and Future Attendance Date Accepted**

A student with date_of_birth = 2030-01-01 is accepted. Attendance for 2027-01-01 is also accepted. Both need date range validation.

4. **BUG-S011: Promote Page - Target Year Shows Only Past Years**

The 'Select Academic Year' dropdown for promotion target only shows 2025-2026 and 2024-2025. Since promotions move students forward, the current/next academic year should be available.

5. **BUG-S012: Subject Attendance API Not Implemented**

The /attendance/subject-attendance/ endpoint returns 404. The UI page exists at /attendance/subject but has no backing REST API. Same for attendance import.

6. **BUG-S013: SMS Sending Time Page Not Implemented**

Page shows placeholder text: 'This Student Info screen is now routed correctly. Legacy parity UI for this page can be implemented next.' Should be either implemented or hidden from navigation.

7. **BUG-S014: Typo - 'Scholership' in Category Dropdown**

The student category dropdown shows 'Scholership' instead of 'Scholarship'. Data-level fix required in admin-setup.

# 5. Improvement Recommendations

## 5.1 Immediate Fixes (Sprint 1)

1. **Implement Permission Middleware:** Create a custom DRF permission class (HasModulePermission) that intercepts every ViewSet action and checks the user role permissions. This is the same fix needed for Admin Section and ALL other modules. Should be implemented ONCE as a global middleware.
2. **Add Input Sanitization:** Strip HTML tags from all text input fields (first_name, last_name, etc.) at the serializer level. Use bleach or django-bleach library. Additionally verify the frontend escapes output properly (React JSX auto-escapes, but check for dangerouslySetInnerHTML usage).
3. **Fix Duplicate Key Error Handling Globally:** Add a global DRF exception handler that catches IntegrityError and returns 400 with a descriptive message. This single fix resolves BUG-S002, S003, S004, and all similar bugs across every module.
4. **Fix Student Export Pagination:** The export function must iterate through ALL API pages or use an unpaginated endpoint. Currently silently dropping 97% of student data.

## 5.2 Short-term Improvements (Sprint 2-3)

1. **Implement Dependent Dropdowns:** When a class is selected, filter the section dropdown to show only sections belonging to that class. Use an API call like /sections/?class_id=X or filter client-side from a class-section mapping object.
2. **Add Date Range Validation:** DOB must be in the past (and reasonable: 3-20 years ago for school students). Attendance date must be today or in the past, within the current academic year. Promotion target year must be current or future.
3. **Implement Soft Delete for Students:** Instead of hard delete (which fails due to FK constraints), implement soft delete (is_active=false, is_deleted=true) with an archive workflow. Students with related data should be archived, not deleted. The 'Disabled Students' page already exists for this purpose.
4. **Register Missing API Endpoints:** Add subject-attendance, attendance-import, and attendance report endpoints to the REST router. Currently the UI pages exist but use internal routes not available via standard API.
5. **Add Server-Side Pagination with Page Controls:** The student list loads all 797 records at once. As the school grows to 2000+ students, this will cause severe performance issues. Implement server-side pagination with page controls (page size, next/previous, total count).
6. **Guardian Module Integration:** The guardian API shows 0 records despite the Add Student form having a guardian section. The form creates guardians but they are not properly linked to the student records via the API. Verify the guardian FK relationship is working.

# 6. AI Integration Strategy

If Claude were building this Student Module to world-class standards, here is how AI would transform it from a basic CRUD system into an intelligent student management platform.

## 6.1 Smart Enrollment & Admission

- **Admission Prediction:** ML model predicts admission likelihood based on inquiry source, class demand, category distribution, and historical conversion rates. Auto-prioritizes high-probability leads for follow-up.
- **Document Verification:** AI-powered OCR extracts data from uploaded birth certificates, transfer certificates, and report cards. Auto-fills student fields (name, DOB, previous school) and flags discrepancies between uploaded documents and manual entry.

## 6.2 Attendance Intelligence

- **Absence Pattern Detection:** AI identifies chronic absenteeism patterns: students with 3+ consecutive absences, declining attendance trends, day-of-week patterns (always absent on Mondays), and seasonal drops. Sends early warning alerts to class teachers and parents.
- **Dropout Risk Scoring:** Combines attendance data with academic performance, fee payment history, and behavioral records to generate a dropout risk score (0-100) for each student. Students above threshold get flagged for counselor intervention.

## 6.3 Academic Performance Prediction

- **Grade Prediction:** Based on attendance patterns, assignment submissions, test scores, and historical data, AI predicts which students are at risk of failing upcoming exams. Provides actionable recommendations: 'Ravi needs 15 extra hours of Math tutoring based on his Unit Test 2 performance.'
- **Learning Gap Analysis:** AI analyzes subject-wise attendance and exam results to identify topic-level gaps. Generates personalized study plans: 'Section 9A has 60% proficiency in Quadratic Equations vs 85% school average - recommend 3 additional sessions.'

## 6.4 Smart Promotion & Placement

- **Promotion Readiness Score:** AI generates a multi-factor promotion readiness score combining attendance (>75% required), exam results (passing in all subjects), behavioral record (no serious incidents), and fee clearance. Automates bulk promotion for qualifying students.
- **Section Balancing:** When promoting students to the next class, AI optimally distributes them across sections to balance gender ratio, academic performance mix, and special needs distribution. Prevents the common problem of one section being significantly stronger than others.

## 6.5 Parent Communication AI

- **Automated SMS/Notification:** AI drafts personalized absence notifications: 'Dear Mr. Reddy, Aarav was absent today (March 28). He has been absent 3 days this month. Please contact the class teacher if he is unwell.' Supports multilingual (English, Hindi, Telugu).
- **Progress Report Generation:** AI generates natural language progress summaries from structured data: 'Priya has shown excellent improvement in Science (from 72% to 88%) but needs attention in Mathematics (declined from 65% to 58%). Her attendance is 92% which is above class average.'

## 6.6 Data Quality & Integrity

- **Duplicate Detection:** Fuzzy matching algorithm identifies potential duplicate students based on name similarity, DOB, parent name, and phone number. Flags 'Aadhya Boya (ADM3542)' and 'Aadhya Boya (ADM3758)' as potential duplicates for review.
- **Data Completeness Scoring:** AI scores each student profile on completeness (0-100%). Currently: blood_group is empty for most students, category is null for many, guardian is null for all 797. Dashboard shows 'Data completeness: 45% - 797 students missing guardian info.'

## 6.7 Cross-Module Intelligence Dashboard

- **Student 360 View:** Single page showing all data for a student: attendance trend chart, exam results timeline, fee payment status, behavioral incidents, transport route, and AI-generated insights. Teachers see this before parent-teacher meetings.
- **Class Health Dashboard:** AI-generated class-level metrics: average attendance, grade distribution, at-risk student count, fee collection rate. Principals see all classes at a glance with red/yellow/green indicators.

# 7. Claude-Level Standard vs Current State

| Area | Current State | Claude Standard |
|---|---|---|
| **Security** | Zero permission enforcement, XSS in names | RBAC + field-level permissions + input sanitization + CSRF + rate limiting |
| **Validation** | Future DOB accepted, HTML in names stored | Age-range DOB, HTML stripped, cross-field validation, real-time feedback |
| **Error Handling** | 500 on duplicates, 500 on FK delete | Global exception handler, structured error codes, graceful FK handling |
| **Export** | Only exports 25 of 797 students | Full export with progress bar, CSV/PDF/Excel, scheduled exports, email delivery |
| **Attendance** | Basic P/A marking only | Face recognition, GPS geofencing, auto-SMS to parents, trend analytics |
| **Dropdowns** | All sections shown for all classes | Dependent cascading dropdowns with AJAX, cached for performance |
| **Search** | Client-side text search only | Server-side full-text, filters by class/section/category/gender/date-range |
| **Guardian** | 0 records linked despite form existing | Full guardian profile, emergency contacts, pick-up authorization, communication log |
| **Promotion** | Manual, target year shows past only | AI-assisted bulk promotion with readiness scoring and section balancing |
| **Reports** | None built-in | Class-wise, section-wise, category-wise, attendance trends, gender analysis, custom date ranges |
| **Mobile** | Responsive only | Parent app for attendance alerts, grade viewing, fee payment, leave application |
| **Data Quality** | No validation on completeness | Profile completeness scoring, duplicate detection, data cleanup campaigns |

# 8. Conclusion

The Student Info module passed 42 of 71 test cases (59% pass rate). The 10 CRITICAL security findings and multiple P1 bugs (XSS vulnerability, broken export, duplicate key errors) make this module unsafe for production use.

The core CRUD operations work correctly for students, categories, and groups. The UI is comprehensive with 15 pages covering enrollment, attendance, promotion, and export workflows. However, the security vulnerability (zero permission enforcement) combined with the XSS vulnerability creates a dangerous combination: any authenticated user can inject malicious scripts into student names that would execute for every admin viewing the student list.

The Student Export bug silently dropping 97% of data is a critical data integrity issue. Schools relying on the export function for backup, reporting, or regulatory compliance would receive incomplete data without any warning.

The AI integration strategy in Section 6 would transform this module from basic data entry into an intelligent student management platform with dropout prediction, attendance intelligence, automated parent communication, and data quality management. The immediate priority must be fixing the security and XSS vulnerabilities.

**Recommended Action:** DO NOT deploy to production until (1) permission enforcement middleware is implemented globally, (2) HTML input sanitization is added, and (3) student export pagination is fixed. All other improvements can follow in subsequent sprints.